

Beyond IID

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Code: <https://github.com/salharayyan1-ui/bias-da-dg-osr>

Compute: single NVIDIA RTX 3060 GPU (PyTorch 2.11, CUDA 12.8), deterministic cuDNN, seed 6304; the reported training runs of Tasks 2–4 took about 5 GPU-hours in total.

Abstract

An ideal representation has to know how to preserve what helps in distinguishing features. Perhaps more importantly, it also needs to know when to abstain. This needs the model to discard noise and learn what it needs to actually generalize well. Around this theme, we have four controlled studies that are cue-bias, domain adaptation and generalization, and open-set rejection. Model backbones lean on different architectural biases, prioritizing visual patterns to make predictions. Forcing source and target domains alignment helps but only in moderation and in extreme cases, destroys class distinguishing features. Sharpness-aware training without seeing target domain at all reaches within 2 points of target-aware training. In open-set recognition, unknowns that the model deems similar to known classes are accepted in majority. Most interestingly, aggregate accuracy numbers across studies differ sometimes greatly from class-specific failures.

Task 1

Dataset/preprocessing: STL-10 dataset comprising 10 classes used. Images 96×96 resized bicubically to 224×224 RGB images. The interventions happened before models’ own normalizations, so this ensures that all models receive the same inputs.

Backbone models evaluated: ResNet-50 pretrained on IMAGENET1K_V2 was used with its 2048-dim pooled feature vector. ViT-B/16 was used which was pretrained on IMAGENET1K_V1 with its 768-dimensional class token. OpenCLIP ViT-B/32 (OpenAI; 512-d normalised embedding) was also evaluated.

Each frozen backbone is equipped with a linear classification head. These were trained on a stratified 80-20 train-validation split with a fixed random seed 6304. The hyperparameters used for training were AdamW optimizer, learning rate 10^{-3} , weight decay of 10^{-4} , maximum 50 epochs with patience of 5 epochs. Additionally, CLIP was evaluated zero-shot using the prompt template “a photo of a {class}”.

For evaluation, a fixed, class-balanced subset of 500 test images was used.

Table 1: Design choices, hypotheses and metrics, stated before the results are interpreted.

Choice	Hypothesis	Metric
Dataset: STL-10	H1: distinct classes give near-equal clean accuracy	clean acc., F1
Colour: grayscale; 180° YIQ hue rotation (luma, edges, saturation kept)	H2: misleading hue hurts more than removed colour	Δ acc., consistency
Cue conflicts: 6 animal–vehicle pairs $\times$ 2 directions, AdaIN $\alpha = 1$	H3: SB < 50% for an ImageNet CNN (Geirhos et al., 2019), not assumed for ViT/CLIP; coverage << 100%	SB with coverage
Projection: t-SNE (cosine, perplexity 30, PCA init, seed 6304) per backbone $\times$ intervention	H4: low normalised stability moves points out of their clean clusters	mixing; norm. I_T

Cue-conflict construction: Total 240 cue-conflict images were produced via Adaptive Instance Normalization (AdaIN; $\alpha=1.0$) across 6 bidirectional animal-vehicle pairs (so 12 total directions). These were airplane-cat, bird-car, dog-ship, deer-truck, horse-airplane, monkey-car, and the same 6 pairs in the opposite direction. Low-contrast/flat outputs ($sd < 0.04$), insufficient style transfers (mean abs pixel difference < 0.05), and shape disruption (edge-map correlation with content image < 0.45) were rejected. From 600 stylization candidates, 503 passed (all 97 rejections were due to the edge rule). The first 20 images per direction were kept. This produced 240 cue conflict images balanced across each direction. All of the 3 models correctly identified the clean content source of these images with accuracies of 97–99%.

Table 2: Clean accuracy and interventions (n = 500; %). Intervention cells: accuracy (Δ versus the model’s own clean accuracy, in points) / prediction consistency with its clean predictions. Conf. = mean maximum softmax probability.

Model	Clean: acc / F1 / conf.	Grayscale	Hue +180°	Patch shuffle 4×4
ResNet-50	98.0 / 98.0 / .976	94.0 (−4.0) / 95.2	95.6 (−2.4) / 96.4	89.2 (−8.8) / 89.6
ViT-B/16	97.8 / 97.8 / .971	96.6 (−1.2) / 97.0	95.6 (−2.2) / 95.8	91.6 (−6.2) / 92.6
CLIP head	98.8 / 98.8 / .964	96.0 (−2.8) / 96.4	96.2 (−2.6) / 96.4	85.0 (−13.8) / 85.2
CLIP zero-shot	97.4 / 97.4 / .951	94.6 (−2.8) / 96.4	95.6 (−1.8) / 96.8	84.0 (−13.4) / 84.0

Table 3: Cue conflicts (n = 240, $\alpha = 1$). Each prediction is shape (content class), texture (style class) or other; SB = $N_{\text{shape}} / (N_{\text{shape}} + N_{\text{texture}})$ with 95% Wilson CI; coverage = $(N_{\text{shape}} + N_{\text{texture}}) / N_{\text{total}}$ (%).

Model	N_{shape}	N_{texture}	N_{other}	SB % [95% CI]	Coverage %	Mean max-prob
ResNet-50	47	44	149	51.6 [41.5, 61.6]	37.9	.855
ViT-B/16	102	23	115	81.6 [73.9, 87.4]	52.1	.637
CLIP head	106	9	125	92.2 [85.8, 95.8]	47.9	.507
CLIP zero-shot	115	14	111	89.1 [82.6, 93.4]	53.8	.578

Findings and research questions

RQ1: Colour, shape, texture biases: All models managed a high accuracy on clean data and were within 1.4 points of each other. Colour serves as a secondary cue which can be seen through grayscale conversion and hue rotations reducing accuracy by only 1.2 to 4.0 points. They alter about 3% to 5% of overall predictions. It is worth mentioning that inverted hue does not cause greater disruption than complete colour removal with only ViT-B/16 showing a slightly higher drop under hue rotation than grayscale. Stylization leaves only 20-48% of images with content labels, so decisions primarily hinge on mid-level statistics more than colour.

In terms of texture and shape bias, ResNet-50 shows a nearly random shape bias at 51.6%, derived from 91 correct decisions ($N_{\text{shape}}=47$, $N_{\text{texture}}=44$). 62.1% of its decisions fall into uninterpretable classes. ResNet does not neatly oscillate between shape and texture. It is fairly confused by the presence of conflicting cues. Still, it produced a huge average maximum confidence score of 0.855 which reminds me of our Dr Tahir’s mention of the term. The model predicts nonsensical classes with a very **high overconfidence**.

ViT-B/16 manages a higher coverage and shows a strong shape bias ($N_{\text{shape}}=102$ vs. $N_{\text{texture}}=23$). Both ViT-B/16 and CLIP achieve significantly higher coverage (52.1% and 47.9%-53.8%) and highly favor shape over texture (81.6% and 92.2%), while maintaining moderate confidence drops (0.637 and 0.507), relative to ResNet-50. Coverage stays below 54% for every model, so each shape bias rests on only 91–129 decisions and is weaker evidence than the percentages suggest.

RQ2: translation and patch shuffling: Spatial translation is rather harmless across the board. Shifting up to 32 pixels (amounting to 14% of frame) changed accuracy by at most 1.4 points and leaves 97.7-99.6% of predictions unchanged (Fig. 1a,b). Shuffling however, shows much more disparity. 4×4 costs ResNet-50 −8.8, ViT-B/16 −6.2 and CLIP −13.8 (head) / −13.4 (zero-shot) points (CLIP’s extra drop is significant against both others). Still, about 84-92% accuracy survived meaning STL-10 is decided from local parts/context. At 8×8, accuracy is 69% (ResNet-50), 81% (ViT-B/16) and 42–49% (CLIP). Clearly, global organisation matters most to CLIP here and least to ViT-B/16.

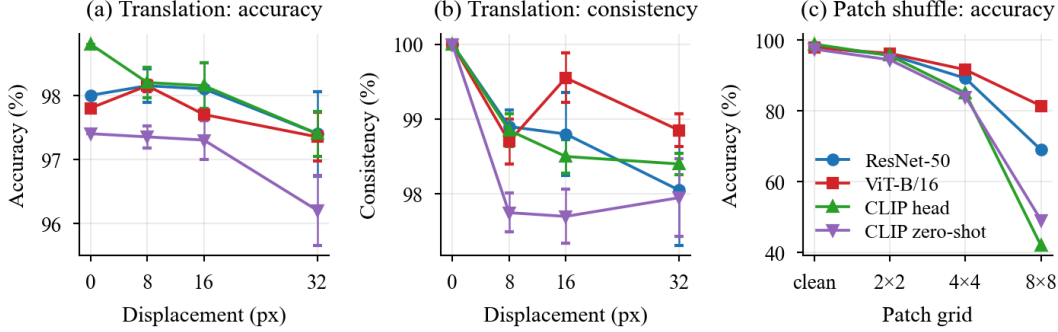


Figure 1: (a, b) Accuracy and prediction consistency versus displacement (0, 8, 16, 32 px; reflection padding, shifted crop; mean $\pm$ s.d. over the four directions). (c) Accuracy versus patch-shuffle grid (one non-identity permutation per image, seed 6304, same images for all models).

Table 4: Representation stability, floor-normalised $(I_T - \text{floor}) / (1 - \text{floor})$ with raw I_T in parentheses (1 = unchanged, 0 = as dissimilar as an unrelated image). Random-pair cosine floors: 0.25 / 0.09 / 0.62.

Intervention	ResNet-50	ViT-B/16	CLIP ViT-B/32
Grayscale	0.72 (0.79)	0.72 (0.75)	0.75 (0.90)
Hue +180°	0.78 (0.83)	0.68 (0.71)	0.75 (0.91)
Cue conflict	-0.05 (0.22)	0.10 (0.18)	0.03 (0.62)
Translation 8/16/32 px	0.96/0.94/0.91	0.94/0.97/0.94	0.95/0.92/0.92
Patch shuffle 4x4	0.62 (0.72)	0.62 (0.66)	0.30 (0.73)

RQ3: predictions and representations: To evaluate how internal features changes relate with final classifications, we first need to calibrate the representation stability across all modes. Unrelated images have very different baseline cosine similarities depending on model backbone so Table 4 normalizes these stability scores against their baseline floors.

Colour interventions moved representations 22-32% towards unrelated images. However, they only affect 3-5% of the predictions, so the backbones preserve colour information that the heads mostly discard during prediction.

Perturbations of the severe type degrade both representation and predictions. Cue conflict intervention brings all three models to near-zero and has quite a large effect on predictions (51-80%) with the t-SNE showing the stylized samples forming a separate, class-mixed cloud in all backbones. Under shuffling, CLIP’s stability and accuracy drops are largest.

Comparing CLIP’s linear head with its zero shot classifier shows that identical feature representations should not be taken to guarantee identical predictions. While both agree on 97.8% of clean images, their agreement is 60.8% with cue conflicts, showing shape bias belongs to representation plus decision rule.

RQ4: architecture and pretraining: To decide whether model behaviors come from architecture or train data, we need to look at the model pairs. ResNet-50 and ViT-B/16 are both trained on ImageNet-1k, use similar data augmentation techniques and achieve nearly the same ImageNet accuracy of about 81%. This means that their behavioral gaps such as ViT’s higher shape bias and ResNet’s larger drop on grayscale images are consistent with architectural differences between convolution and self-attention. However, I must mention that ViT has 86.6M params and ResNet has 25.6M params and there are some differences in train schedule and resolution so architecture still remains only a strong hypothesis.

Comparing CLIP ViT-B/32 and ViT-B/16 mostly isolates non architectural factors since both are ViTs, apart from patch size. Despite this, CLIP displays a higher shape bias and collapses more badly under patch shuffling. These differences likely come from CLIP’s multimodal pretraining on the image-text pairs and/or its larger 32px patch size, which cannot be separated here.

Additionally, I have learned from my prior research projects that a single seed is never enough to produce concrete statements for anything really.

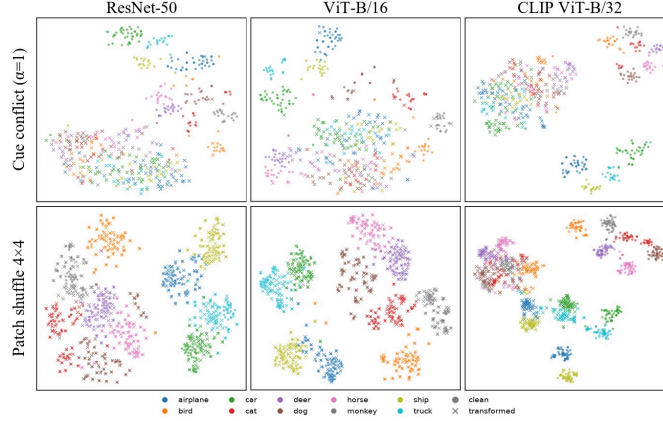


Figure 2: t-SNE of clean (•) and transformed (×) features, colour = class (content class for cue conflicts); cosine metric, perplexity 30, PCA init, seed 6304, one fit per panel (coordinates not comparable across panels).



Figure 3: Cue-conflict examples (content | style | stylised) with predictions of ResNet-50 (R50), ViT-B/16, CLIP head (-h) and zero-shot (-zs), coloured **shape**, **texture**, other. (a) All follow shape. (b) Only ResNet-50 follows texture. (c) Four different answers. (d) Shape barely visible; every answer is “other”.

Task 2

Dataset and domains: PACS dataset used. It has 7 object categories in total across four different domains. The source domains are Photo, Art Painting, and Cartoon with labels provided. Each domain is split 80/20 into train/val sets using the same seed 6304. The target domain is Sketch images and they do not have the labels. Ground truth labels for Sketch were only used for the final test eval after all training concluded.

Model and train information: Base network is a ResNet-18 pretrained on ImageNet with a final 7-class output head. The entire network is fine-tuned with BatchNorm running averages frozen at the ImageNet values. The images were resized to 256×256 , then randomly cropped to 224×224 and randomly flipped horizontally. The optimizer used was AdamW with a $lr=10^{-4}$, patience=5, and weight decay of 10^{-4} , running for up to 30 epochs with seed 6304 reset before each run. Each step paired 24 source domain images (8 from each source domain) with 24 Sketch target domain images, executing 234 updates per epoch.

Methods:

Source-only: standard ERM CrossEntropy over pooled source train split

Deep Adaptation Network: Kernel Maximum Mean Discrepancy with alignment weight=1, using multi-kernel combination of 3 gaussian RBF kernels scaled at $0.5\times$, $1.0\times$, $2.0\times$ the median pairwise squared batch distance.

Domain-Adversarial Neural Network: Adversarial domain discriminator (256 hidden units, dropout 0.5) connecting using a GRL (gradient reversal layer) with dynamic schedule ($\alpha(p) = 2/(1 + e^{-10p}) - 1$, unit weight)

Conditional Domain Adversarial Network: 256 hidden unit discriminator conditioned on multilinear representation vec with no detached gradients, no entropy weighting.

Deviations in technical implementation: I adopted the unbiased U-statistic for MMD² in DAN because the diagonal terms in standard biased version drove representations to collapse in initial testing. The math behind the collapse is intriguing.

$$\text{MMD}_{\text{biased}}^2 = \frac{1}{m^2} \sum_{i=1}^m \sum_{j=1}^m k(x_i, x_j) + \frac{1}{n^2} \sum_{i=1}^n \sum_{j=1}^n k(y_i, y_j) - \frac{2}{mn} \sum_{i=1}^m \sum_{j=1}^n k(x_i, y_j)$$

The Gaussian kernel evaluated at diagonal self-similarity terms ($k(x_i, x_i) = 1$) introduces a strictly positive sample bias of $\frac{1}{m} + \frac{1}{n}$. Under frozen BatchNorm, gradients drove representations toward an all-zero collapse to minimize this term. The fix was removing the diagonal self-comparisons using the unbiased U-statistic:

$$\text{MMD}_{\text{unbiased}}^2 = \frac{1}{m(m-1)} \sum_{i=1}^m \sum_{j \neq i}^m k(x_i, x_j) + \frac{1}{n(n-1)} \sum_{i=1}^n \sum_{j \neq i}^n k(y_i, y_j) - \frac{2}{mn} \sum_{i=1}^m \sum_{j=1}^n k(x_i, y_j)$$

Without norm constraints, DANN diverged in the first epoch as feature norms grew roughly $1.5\times$ per update step, forcing me to constrain and scale discriminator inputs (to $\sqrt{512} \cdot f / \|f\|_2$). Worth mentioning here that in my first run, I used raw unit-norm vectors ($f / \|f\|_2$), but this actually led the opposite failure mode as the input logits were pinned to near zero, neutering the discriminator. Multiplying the unit norm vectors by $512^{0.5}$ stabilized the discriminator but it reduced DANN’s Sketch accuracy greatly while increasing CDAN’s accuracy slightly. This change was motivated by the discriminator’s near-zero logits, but the first run’s Sketch results had already been seen, which is a limitation.

Table 5: Source-only, DAN, DANN and CDAN (%), one run each). Source val. = the three source validation splits (Art is the worst source for every method); Sketch = all 3,929 images, labels used only after every checkpoint was fixed; Δacc = Sketch accuracy minus Source-only’s (paired-bootstrap 95% CI); domain sep. = held-out accuracy of a class-balanced logistic-regression probe ($C = 1$, standardised features) separating 1,212 source-validation from 1,212 Sketch features (728 test images; chance 50%).

Method	Source val. accuracy			Source val. mean		Sketch (target)			Domain sep.
	Photo	Art	Cartoon	acc	F1	acc	F1	Δacc [95% CI]	
Source-only	96.1	93.4	94.7	94.7	94.3	57.2	60.2	–	99.7
DAN	96.7	89.3	93.0	93.0	92.3	75.8	73.2	+18.6 [+16.8, +20.5]	93.0
DANN	95.8	94.9	94.9	95.2	94.9	24.8	13.3	–32.4 [–34.2, –30.6]	99.9
CDAN	95.8	91.0	93.8	93.5	93.4	66.5	58.1	+9.3 [+7.6, +11.0]	99.2

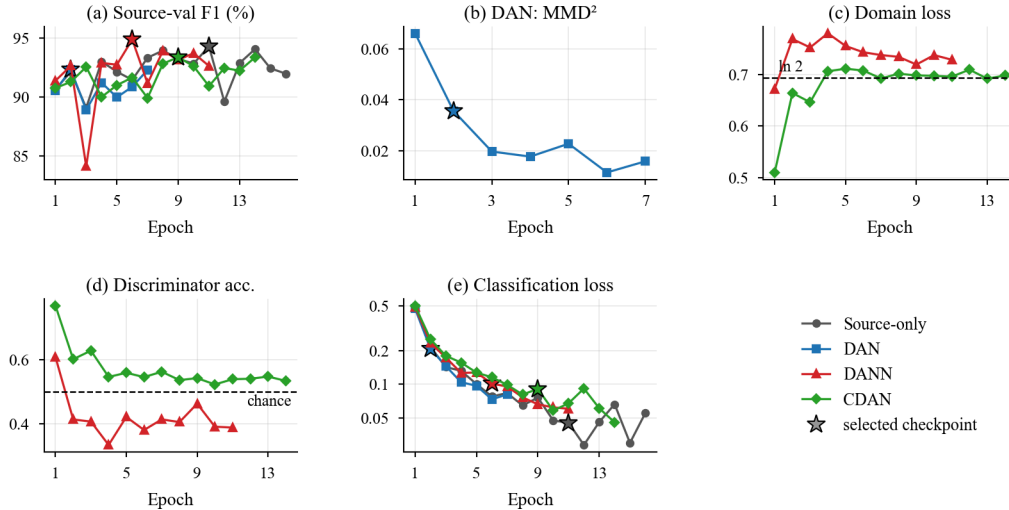


Figure 4: Did each method train as intended? (a) Mean source-validation macro-F1 per epoch (★ = selected checkpoint). (b) DAN’s MMD^2 falls from 0.066 to 0.016. (c, d) DANN’s and CDAN’s domain loss (dashed: $\ln 2$) and discriminator accuracy (dashed: chance). (e) Training classification loss falls from 0.47–0.50 to 0.05–0.08 for all four methods.

Findings and research questions

RQ1: the domain gap and its failing classes: The baseline model scored 94.7% accuracy on source images but plunged to 57.2% on sketches. Two animal classes cause most (2/3rd) of the failures here (Horse and Dog). Errors do not scatter randomly with 56% of real horses and 24% of

real dogs labelled as giraffes while 27% of dogs labelled as elephants. In fact, giraffes and elephants make up 38.5% and 25.8% of all model predictions, even though they are actually 19.2% and 18.8% of true target images. Clearly, the model, when confused by an animal sketch, dumps its prediction mostly onto giraffes and elephants class. Interestingly, man made objects hold up well here. House recall drops by 6.4 points compared to horse recall which dropped by 74.6 points. The problem is further complicated by class imbalance between domains. House and Person make up 35.5% of all source domains but only 6.1% of the Sketch domain.

RQ2 does lower separability mean better recognition?: At first look, lower separability seems to align with higher test accuracy. DAN has lowest separability (93.0%) and highest accuracy (75.8%), followed by CDAN (99.2% / 66.5%), Source-only (99.7% / 57.2%), and DANN (99.9% / 24.8%). This is weak evidence however. The linear probe separates Source-only, CDAN and DANN at $\geq 99.2\%$ accuracies, but their Sketch accuracies are much different with a difference of up to 41.7 points. Inside DAN’s weight tests, separability drops steadily as alignment weight increases, but Sketch accuracy peaks then worsens. In my case, separability worked well as a red flag for failure. DANN’s separability was just as high as the baseline, showing it failed to blend the domains. Class-level evidence shows trade-offs well (Fig. 5a). DAN raises accuracy by 18.6 points and macro-F1 by 13.0 but loses 11.6 points on elephants. DAN achieved 93.8% recall on person, but with a precision of only 36.2% due to overprediction. CDAN’s accuracy gain (+9.3) comes with a macro-F1 change of -2.1 . DANN is negative transfer in six of seven classes (net $-1,274$ images). It predicts giraffes for 85.3% of Sketch, and 132 of its 179 fixes are giraffes.

RQ3 conditional versus marginal alignment: Conditioning stopped the model from crashing. In DANN, discriminator broke down and accuracy went to 24.8%. In CDAN, the class predictions kept the discriminator training stable. So, CDAN beat DANN by 41.7 points difference. Here conditioning clearly helped.

However, simple feature mixing still won. DAN beat CDAN by 9.3 points. To explain this, consider the metaphor ‘**blind leading the blind**’. This is likely what happened as CDAN relies on its own predictions to figure out which class the unlabelled Sketch comes from. These guesses are unreliable and sometimes even align the wrong things. For example, if the model mistakenly thinks a Sketch image is of a dog, it will actively teach itself incorrect associations with no way to correct itself since there are no target domain labels. In fact, **82% of sketched people and 6–9% of all other classes were misclassified as house**, causing house to swallow 13.0% of all target predictions (true share: 2.0%; house precision: 15.7%). This failure was further amplified by dataset imbalances (houses are 14.2% of source data but only 2.0% of Sketch). In source domain data, houses and people are 35.5% but in Sketch, they are only 6.1%. When CDAN tries matching class by class, distributions across domains are different and forcing features to line up messes up the representations further. So conditioning in this PA helped training stability and accuracy relative to DANN, but not relative to DAN.

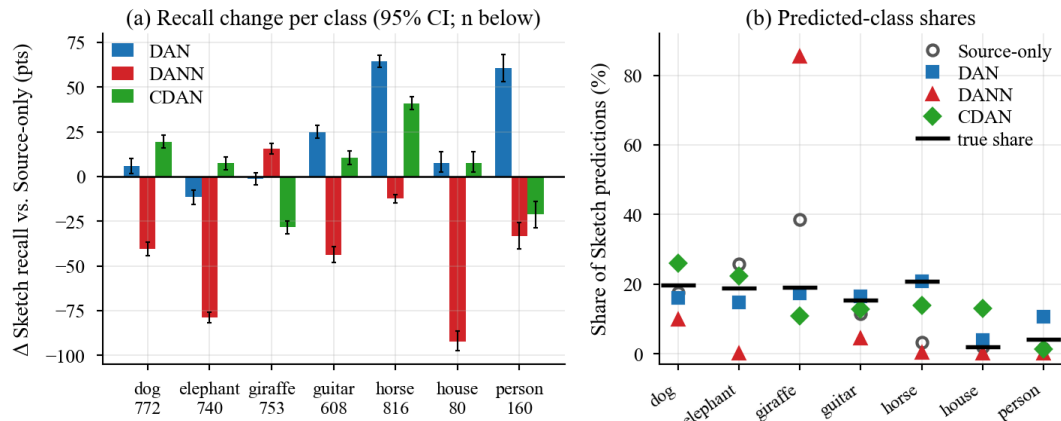


Figure 5: (a) Change in Sketch recall relative to Source-only, per class, with paired-bootstrap 95% CIs (house $n = 80$, person $n = 160$). (b) Share of Sketch predictions per class against the true share.

RQ4 alignment strength and label-free selection:

Stronger alignment reduces overall source accuracy (94.7% to 89.7%). Sketch accuracy increased but then inverted. For DAN, at $\lambda = 10$ both source and target decline relative to $\lambda = 1$. At the sweet spot $\lambda = 1$, a modest 1.7-point drop in mean source accuracy (borne mainly by Art Painting, down 4.1 points from 93.4% to 89.3%) yields an optimal +18.6 point gain on Sketch, representing a worthy trade-off. If we pick the highest source validation score (mean source-validation macro-F1), we'll get $\lambda = 0.1$ which would miss the best model at $\lambda = 1.0$. If we choose the model with lowest separability, we'll get $\lambda = 10$ which is actually the worst model because of over-alignment. Across this task's eight runs (two of them weaker-reversal DANN runs not shown here), source-validation macro-F1 is negatively correlated with Sketch accuracy ($\rho = -0.71$), and the collapsed DANN apparently has the highest source F1 (94.9%) despite having the worst Sketch accuracy. Instead, label-free model health must rely on structural red flags like discriminator accuracy falling below chance (< 0.50), probe separability failing to drop below the baseline, and target predictions collapsing onto a single dominant class.

Table 6: Controlled design study: DAN with $\lambda_{\text{MMD}} \in \{0.1, 1, 10\}$, all else fixed (%). Expectation stated before any target result: source validation flat or slightly lower, separability lower, and Sketch accuracy higher at moderate λ but lower at large λ if alignment starts to mix classes.

λ_{MMD}	Source acc	Source F1	Art acc (worst)	Domain sep.	Sketch acc	Sketch F1	Δ Sketch acc [95% CI]
0 (Source-only)	94.7	94.3	93.4	99.7	57.2	60.2	—
0.1	94.2	93.7	92.4	97.4	72.2	67.4	+15.0 [+13.3, +16.6]
1 (main)	93.0	92.3	89.3	93.0	75.8	73.2	+18.6 [+16.8, +20.5]
10	89.7	89.4	87.8	79.9	60.6	56.7	+3.4 [+1.3, +5.4]

Task 3: Domain generalization on PACS (unseen target: Sketch)

Setup, dataset, models, techniques and training: The baseline protocol utilizes the same PACS dataset as Task 2 with the same stratified 80/20 splits for the labeled sources, same architecture, seed, preprocessing steps, AdamW optimizer, early stopping configuration, and checkpoint selection. The **ERM** baseline is Task 2's Source-only checkpoint. Furthermore, **Deep Adaptation Network for Domain Generalization (DAN-DG)** incorporates the unbiased MMD² U-statistic and three RBF kernels. The target Sketch domain is strictly unseen during training. Updates contain 8 images per source domain only, DAN-DG applies its loss as λ_{DG} times the mean MMD² across three source pairs. **Sharpness-Aware Minimization (SAM)** seeks flat loss minima by minimizing worst-case loss within $\rho = 0.05$ around the weights with frozen BatchNorm statistics.

Domain separability is the held-out accuracy (chance 33.3%) of a logistic-regression probe ($C = 1$, standardized features). Δ_{sharp} is the cross-entropy increase after one normalized gradient-ascent step of radius 0.05 on a fixed batch of 32 validation images per source. The controlled study is the λ_{DG} one (SAM's ρ was not varied).

Table 7: ERM, DAN-DG and SAM (% , one run each). Each source cell is accuracy / macro-F1 on that domain's validation split (333 / 410 / 469 images); Mean is over the three domains and Worst is the lowest domain (Art Painting for all three models). Sketch = all 3,929 images, evaluated after every setting was locked; Δ acc = Sketch accuracy minus ERM's (paired-bootstrap 95% CI).

Method	Source validation: accuracy / macro-F1					Sketch (target)		
	Photo	Art	Cartoon	Mean	Worst	acc	F1	Δ acc [95% CI]
ERM	96.1 / 95.1	93.4 / 93.0	94.7 / 94.7	94.7 / 94.3	93.4 / 93.0	57.2	60.2	—
DAN-DG	97.6 / 97.1	92.0 / 91.8	94.0 / 94.2	94.5 / 94.4	92.0 / 91.8	58.5	63.6	+1.3 [−0.8, +3.4]
SAM	96.4 / 95.5	94.9 / 94.9	95.7 / 96.1	95.7 / 95.5	94.9 / 94.9	74.2	76.0	+17.0 [+15.3, +18.7]

RQ1) do source-side scores predict Sketch, and what do they hide?: Source validation correctly identifies the best model but fails on close comparisons. SAM achieved the highest source performance across the board except Photo (95.7% accuracy, 95.5% macro-F1, and 94.9% worst-domain F1 on Art Painting), actually reflecting the highest target performance on Sketch (74.2% accuracy, +17.0 points over ERM [95% CI: +15.3, +18.7]; 76.0% macro-F1, +15.8 points over ERM). Between ERM and DAN-DG, however, the source validation metrics almost tie on mean source F1 (94.4% vs. 94.3%) and DAN-DG slightly underperforms on the hardest source domain (91.8% vs. 93.0% on Art Painting). Still, DAN-DG outperforms ERM on Sketch by 1.3 accuracy points and 3.4 macro-F1 points. These small margins fall within random noise (domain standard error is 0.8–1.3 points) and make it hard to distinguish models reliably. The exception is

the DAN-DG macro-F1 gain (95% CI [+1.5, +5.3]). All runs use one seed, so the CIs cover only Sketch image sampling.

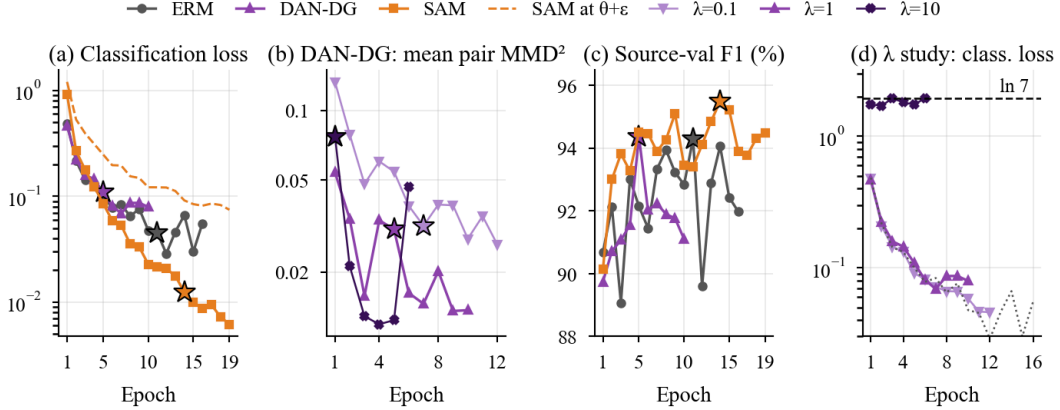


Figure 6: Did each method train as intended? (a) Training classification loss per epoch (log scale; dashed: SAM’s loss at the perturbed point $\theta+\epsilon$); $\star$ = selected checkpoint. (b) Unweighted mean pairwise MMD² of DAN-DG for the three λ_{DG} (the loss adds λ_{DG} times this). (c) Mean source-validation macro-F1. (d) Classification loss of the λ_{DG} study (dotted: ERM; dashed: $\ln 7$, a uniform prediction).

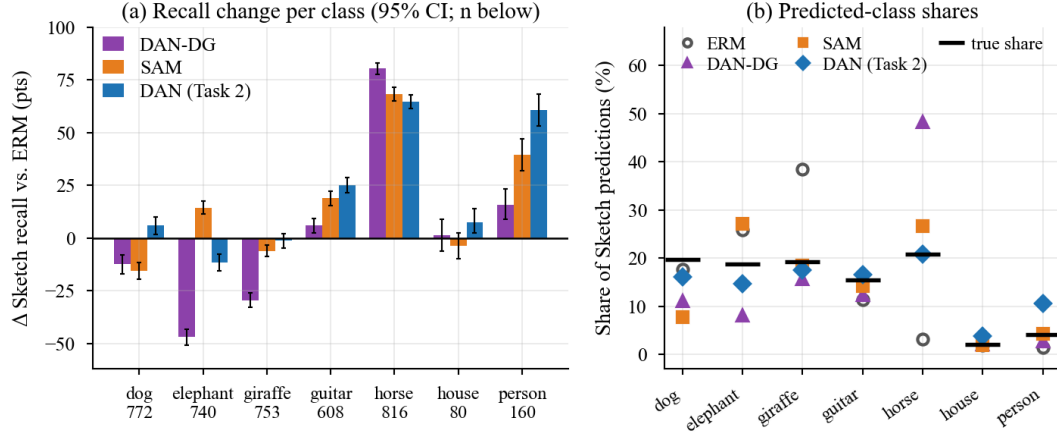


Figure 7: Per-class Sketch recall changes relative to ERM and predicted vs. true class shares. (a) Change in Sketch recall relative to ERM per class with paired-bootstrap 95% CIs, for DAN-DG, SAM and the target-aware DAN of Task 2 (house $n = 80$, person $n = 160$). (b) Share of Sketch predictions per class against the true share.

Overall averages also mask extreme class-level imbalances. DAN-DG’s net gain of 51 images (917 fixed, 866 broken) relied almost entirely on fixing horses (+655 images), with minor gains in guitars (+36), persons (+25), and houses (+1). This boost in horses caused other animals to drop, elephants losing 348 images (recall drops from 79.3% to 32.3%), giraffes 222, and dogs 96. SAM’s net gain of 669 images also comes at the cost of losing 121 correct dog predictions (recall dropped from 48.7% to 33.0%). Horses are the baseline’s main failure on Sketch (recall 13.5%), despite all models scoring at least 91.8% on every source domain.

RQ2) does DAN-DG make the sources less distinguishable, and does that help?: DAN-DG reduces domain separability without total elimination. Source separability drops from 86.3% in ERM to 71.7% (well above the 33.3% random baseline) in DAN-DG ($\lambda_{DG} = 1$). SAM barely affects domain separability (at 84.3%). Lower domain separability does not automatically guarantee better target accuracy. The λ_{DG} sweep shows separability barely moves at $\lambda_{DG} = 0.1$ (86.7%). This setting produced the highest Sketch accuracy in the sweep at 60.6% (+3.4 points over ERM; 95% CI: [+1.8, +5.1]; macro-F1 +0.4). At $\lambda_{DG} = 1$, a large 14.7-point drop in separability (down to 71.7%) produces only a small 1.3-point Sketch gain (58.5% accuracy; 95% CI: [−0.8, +3.4]; macro-F1 +3.4). Pushing alignment to $\lambda_{DG} = 10$ causes source accuracy to crash to 43.5% and Sketch accuracy to 22.7% (−34.5 points; 95% CI: [−36.5, −32.5]).

Table 8: Controlled hyperparameter study of DAN-DG across lambda values (0.1, 1, 10) compared to ERM and SAM, showing source accuracy, source F1, worst-domain F1, domain separability, sharpness proxy Δ_{sharp} , Sketch accuracy, Sketch F1, and Sketch accuracy differences vs ERM with 95% CIs. All values in %; source columns are means over the three validation splits; Δ_{sharp} at $\rho = 0.05$; chance separability 33.3%.

λ_{DG}	Source acc	Source F1	Worst F1	Domain sep.	Δ_{sharp}	Sketch acc	Sketch F1	Δ Sketch acc [95% CI]
0 (ERM)	94.7	94.3	93.0	86.3	0.318	57.2	60.2	—
0.1	94.0	94.0	93.1	86.7	0.353	60.6	60.6	+3.4 [+1.8, +5.1]
1 (main)	94.5	94.4	91.8	71.7	0.300	58.5	63.6	+1.3 [−0.8, +3.4]
10	43.5	37.8	28.9	68.7	2.274	22.7	24.6	−34.5 [−36.5, −32.5]
SAM, $\rho = 0.05$	95.7	95.5	94.9	84.3	0.091	74.2	76.0	+17.0 [+15.3, +18.7]

This happens because alignment ditches class-discriminative information. At $\lambda_{\text{DG}} = 10$, collapse is visible directly on the sources, with class probe accuracy falling from 95.3% to 56.2%, feature rank dropping from 11.7 to 2.7, and loss stalling near random guessing (1.70–1.94 vs. $\ln 7 \approx 1.95$). At $\lambda_{\text{DG}} = 1$, source metrics look intact (94.1% probe), but target animals are dumped into horse class as shown in Fig. 7b. Whether this comes from alignment mixing class structure or from Sketch strokes interacting with the aligned features is not settled by these runs.

RQ3) does SAM reduce the sharpness proxy, and does that ranking match Sketch?: Yes. SAM’s sharpness proxy (0.091) was about 3.3–3.5 $\times$ flatter than ERM (0.318) and DAN-DG (0.300), and it remained consistently lowest across every perturbation radius tested. Sharpness ranking mirrors Sketch accuracy, but only meaningfully for SAM. The flatness ranking (SAM < DAN-DG < ERM) reflects exactly in Sketch accuracy (74.2% > 58.5% > 57.2%). However, this match is only meaningful for SAM as ERM and DAN-DG differ by only 6% at $\rho = 0.05$ and actually swap order at $\rho = 0.1$, putting them within noise of each other. The proxy uses one fixed batch of 96 validation images, and SAM also trained longer and has higher source scores, so its +17.0 Sketch points cannot be credited to flatness alone.

RQ4) what do unlabeled Sketch images add (Task 2 DAN vs. DAN-DG)?: Unlabeled target data access provided an advantage to DAN outperforming DAN-DG by 17.3 accuracy points. However, it is worth stating here that target data is not an absolute necessity. SAM misses DAN only by 1.6 accuracy points and surpasses it in macro-F1 by 2.7 points. Lastly, DAN and DAN-DG differ in other areas like batch sampling, checkpoint timing etc., so the performance gap shouldn’t be credited entirely to target data access.

Task 4: Open-set recognition (CIFAR-10 known, CIFAR-100 unknown)

Setup and model training: Open-set recognition (OSR) requires classifiers to correctly classify known training classes while rejecting unseen categories. Known dataset comprises the 10 classes in CIFAR-10 (stratified 90%-10% train-val partition, seed 6304) and unknown dataset is made from CIFAR-100 reserved for final testing. It has near unknowns which are similar in visuals to CIFAR-10 classes and far unknowns which are semantically different from CIFAR-10 classes. Each unknown class has 100 images. The model used is ResNet-18 modified for CIFAR-10 dims. Four model configurations were tested. A Vanilla baseline optimized with SGD over 100 epochs (initial learning rate 0.1 with cosine decay). GCSC, which adds RandAugment (two operations at magnitude 9). PROSER, which fine-tunes Vanilla for 50 epochs at a learning rate of 10^{-3} using 5 dummy classes, splitting batches between classifier placeholders ($\beta = 1$) and manifold mixup layer-2 feature interpolation ($\gamma = 0.1$, Beta(2, 2)). An untuned RPL implementation also exists that pushes class representations away from reciprocal points. Every scoring function produces a scalar unknownness score. A higher unknownness means greater chance of sample being from unknown dataset. Evaluated scoring functions include maximum softmax probability (MSP: $1 - \max p$), maximum logit score (MLS: $-\max z$), energy pooling ($-\log \sum \exp(z)$), minimum Mahalanobis distance to class centroids using a shared diagonal covariance, PROSER’s dummy-minus-known logit difference, and RPL distance metrics. Also, an operational threshold τ is set at the 95th percentile of validation anomaly scores.

Table 9: Post-hoc scores on the frozen Vanilla model. AUROC: the 10,000 known test images against the near, far or all unknowns (800, 800, 1,600). Rejection: share of unknowns above τ , the 95th percentile of the score on the CIFAR-10 validation images. Known acc.: share of known test images accepted at τ (the aim is 95%).

Score (Vanilla)	AUROC, known test vs.			Rejection (%) of			Known acc. at τ (%)
	near	far	all	near	far	all	
MSP	0.821	0.900	0.861	30.8	47.0	38.9	94.77
MLS	0.806	0.899	0.853	34.3	56.9	45.6	94.74
Energy	0.806	0.900	0.853	33.3	57.4	45.3	94.91
Mahalanobis	0.809	0.921	0.865	29.6	54.5	42.1	94.68

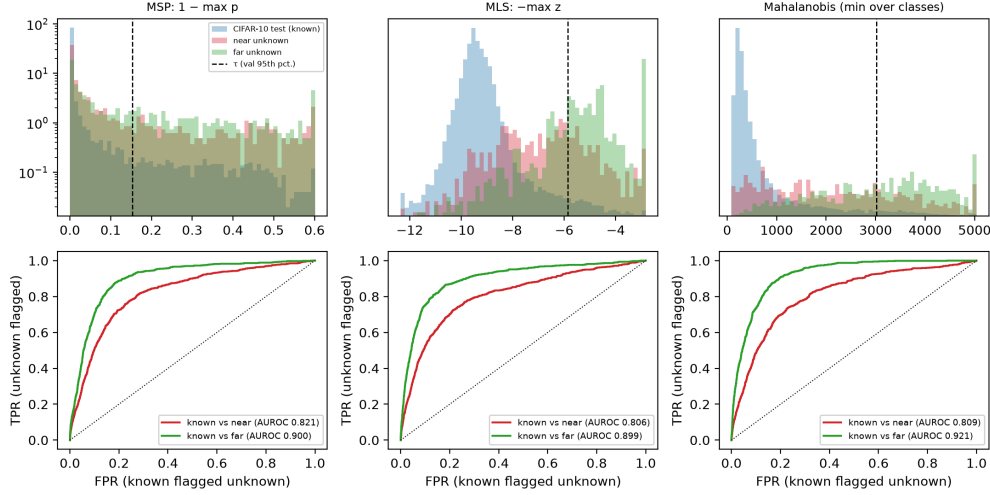


Figure 8: Vanilla score distributions and ROC curves. Top: score distributions (density; log scale for MSP) of CIFAR-10 test images (blue), near (red) and far (green) unknowns; the dashed line is τ . Bottom: ROC curves, known vs. near (red) and known vs. far (green); the AUROCs are in Table 9. Scores beyond the 0.5th and 99.5th percentile of all scores are piled into the end bins, which makes the spikes at the right edges.

Detection effectiveness on Vanilla baseline depended on scoring metric and unknown type. MSP ranks near unknowns best with an AUROC of 0.821 (versus 0.806–0.809 for other scores). But it rejects the fewest far unknowns. Mahalanobis leads in ranking far unknowns (0.921 AUROC). But it rejects fewer total unknowns than MLS. Known test acceptance remains consistently close to the 95% target across all scores, spanning 94.68% to 94.91%.

RQ1) how does similarity to a known class affect rejection?:

Table 10: Acceptance rate per unknown class and assigned CIFAR-10 labels. Vanilla + MLS at $\tau = -5.86$: share of each unknown class accepted (100 images each) and the CIFAR-10 labels given to the accepted images (image count; top three).

Near unknowns			Far unknowns		
Class	Accepted	Labeled as (count)	Class	Accepted	Labeled as (count)
pickup truck	91%	automobile 61, truck 28, airplane 2	bowl	52%	cat 14, frog 10, automobile 8
bus	86%	truck 70, automobile 14, ship 1	wardrobe	52%	cat 30, truck 21, horse 1
wolf	78%	cat 37, dog 37, deer 2	mushroom	46%	bird 19, cat 10, frog 8
fox	71%	cat 31, deer 22, dog 12	chair	43%	airplane 18, bird 11, cat 7
leopard	55%	cat 24, frog 21, deer 4	clock	43%	cat 14, automobile 9, ship 7
tractor	54%	truck 47, ship 4, automobile 3	sunflower	42%	bird 16, dog 11, frog 6
camel	52%	deer 16, dog 12, bird 10	bottle	34%	cat 13, ship 9, truck 6
motorcycle	39%	automobile 24, truck 8, airplane 5	keyboard	33%	airplane 10, bird 7, truck 7
all 8 near	65.8%	truck 153, automobile 102, cat 96 (of 526)	all 8 far	43.1%	cat 99, bird 65, truck 48 (of 345)

Visual similarity weakens unknown detection. Vanilla MLS model accepts 65.8% of near unknowns vs 43.1% of far unknowns. Classes with direct links to CIFAR-10 labels suffer highest error rates. Such inputs are absorbed by the counterpart categories in CIFAR-10. Far unknowns are less sneaky and end up in erratic labels. “CAT” in CIFAR-10 receives the most of all accepted far unknowns. These errors are likely initiated by background cues like sky or walls instead of the object itself.

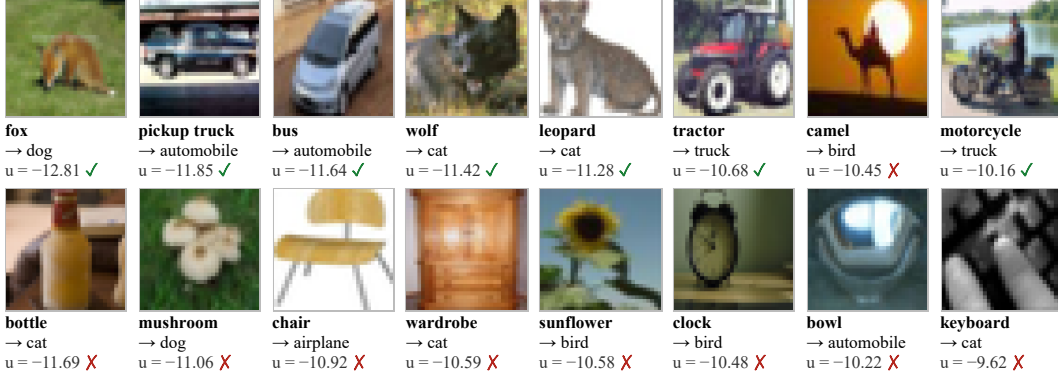


Figure 9: Confidently accepted images for each unknown class. The most confidently accepted image of each unknown class under Vanilla + MLS (top: near, bottom: far), i.e. the worst case of each class, not a typical one. Each label gives the true class $\rightarrow$ the predicted CIFAR-10 class and the score u ; all are accepted because $u \leq \tau = -5.86$ (lower u = more confident). ✓ plausible confusion (the predicted class is a close relative), ✗ surprising; the marks are a subjective judgement.

RQ2) what do MSP, MLS, Energy and Mahalanobis each capture?:

MSP measures predictive confidence through softmax sharpness. MLS tracks the magnitude of the strongest raw logit. Energy aggregates total logit volume. Mahalanobis examines feature distances to class centers.

MLS and Energy are almost identical, and differ on only 42/1600 unknown test images. MSP is aligned with both but Mahalanobis provides an independent geometric view that catches far unknowns which logit-based scoring loses. As Table 9 shows, the optimal score depends on the unknown group and the eval metric. MSP ranks best near unknowns and Mahalanobis on far unknowns. AUROC doesn't neatly reflect rejection. Mahalanobis achieved the higher AUROC but it was MLS that rejected the larger share of both near and far unknowns at the operational 95% threshold.

RQ3) do GCSC's random augmentations improve closed-set accuracy, rejection, both or neither?: RandAug provides a slight benefit only. As Table 11 shows, closed-set accuracy rose slightly by 0.27 points, which may as well be explained by random standard error. Detection ranking remains unchanged with AUROC changes for MLS also within random noise. GCSC scored a minor improvement in rejection at the 95% threshold for both near and far unknowns. This possibly is owed to how data augmentation does not reveal to the model any unseen distributions (i.e., it cannot birth the novelty needed). Thus, semantically similar unknowns still end up getting mapped to known categories.

RQ4) does PROSER improve rejection beyond Vanilla and GCSC, and what does it cost in CSA?:

Table 11: Model comparison across CSA, AUROC, and rejection. CSA is closed-set accuracy on the 10,000 test images, computed from the ten known logits. MLS is the common score; the placeholder row uses $u =$ largest dummy logit – largest known logit. Below the rule (supplementary): the untuned RPL with its two scores, $1 - \max \text{softmax}(d)$ and $-\max d$, where d are the distances to the reciprocal points. One run per model; differences in the text use unrounded values.

Model · score	CSA (%)	AUROC, known test vs.			Rejection (%) of			Known acc. at τ (%)
		near	far	all	near	far	all	
Vanilla · MLS	94.94	0.806	0.899	0.853	34.3	56.9	45.6	94.74
GCSC · MLS	95.21	0.814	0.900	0.857	38.6	59.1	48.9	94.60
PROSER · MLS	94.74	0.812	0.915	0.864	34.0	57.4	45.7	94.57
PROSER · placeholder score	94.74	0.797	0.907	0.852	31.8	53.0	42.4	94.45
RPL · $1 - \max \text{softmax}(d)$	94.90	0.760	0.883	0.821	32.5	49.1	40.8	94.49
RPL · $-\max d$	94.90	0.582	0.639	0.610	25.1	35.9	30.5	94.45

Nope. PROSER did not offer an advantage over GCSC or even Vanilla. As Table 11 shows, it hit a slight ranking gain in far unknowns with its rejection rates at the operational threshold remaining unchanged and behind GCSC. As for cost in closed-set performance, PROSER's CSA drops 0.2 points against Vanilla and 0.47 against GCSC. Underlying mechanics can explain why placeholder

training fails with near unknowns. Even though the model learns to send synthetic mixed features to dummy labels, Manifold Mixup only populates the empty space in between different categories. Near unknowns already resemble some specific known class meaning they end up within or alongside a known class. So the boundary interpolation may push away some far unknowns, but it leaves near-unknown detection mostly untouched. RPL, which learns what each class is not, does not help either: it is below Vanilla, GCSC and PROSER in every AUROC and rejection column of Table 11 (RPL is untuned).

Cross-task discussion and conclusion

Cues and shift: Little shifts in color and translation tend to hurt very less as compared to structural and style shifts like patch scrambling which caused severe drops in accuracy. On the PACS, performance went from approximately 95% baseline to about 57% with most of the errors sinking into a few classes only. This sinking of predictions to few classes was observed in Task 1 as well where cue conflicts routinely sent predictions to nonsensical classes, sometimes with great confidence.

Invariance and discriminability: Forcing source and target domain representations to look similar cannot be taken to guarantee better classification results. Moderate alignments helped in Task 2 when target data was present but evidence shows how alignments of the extreme type tend to mess up class-defining features, leading to class collapses regardless of whether target domain data was used or not. That said, SAM, without ever seeing target domain, ended up very close (within 2 points) to DAN which had ample access to target domain examples.

Recognition and rejection: High classification accuracy does not automatically mean safe open-set rejection. GCSC and PROSER both helped rejection just a little. Near unknowns passed through about two-thirds of the time, usually into a related counterpart label in the known domain. As with Task 1’s ResNet-50, confidence is not the evidence of learning.

Synthesis: A representation that is actually reliable keeps features that define a class well, while ignoring minor distractions. This means avoiding mixing up different categories and guessing when an unfamiliar image looks kind of like a known one. Common shortcut metrics like source-validation accuracy, domain separation, high model confidence, among others, can give false sense of success. In fact, overall average scores tend to hide major failures that pop up when checking particular classes separately.

On a personal note, I ran into far more failure modes during this assignment than I expected. That stood in contrast to how these training methods felt to me in class, almost as if they were bulletproof new ways of training models I had never thought of. It brought to mind something else we’d discussed: there are no incorrect results, only incorrect interpretations.

References

- Chen, G., Qiao, L., Shi, Y., Peng, P., Li, J., Huang, T., Pu, S., and Tian, Y. (2020). Learning open set network with discriminative reciprocal points. In *European Conference on Computer Vision (ECCV)*, pages 507–522.
- Coates, A., Ng, A., and Lee, H. (2011). An analysis of single-layer networks in unsupervised feature learning. In *International Conference on Artificial Intelligence and Statistics (AISTATS)*, pages 215–223.
- Cubuk, E. D., Zoph, B., Shlens, J., and Le, Q. V. (2020). RandAugment: Practical automated data augmentation with a reduced search space. In *CVPR Workshops*, pages 702–703.
- Dosovitskiy, A., Beyer, L., Kolesnikov, A., Weissenborn, D., Zhai, X., Unterthiner, T., Dehghani, M., Minderer, M., Heigold, G., Gelly, S., Uszkoreit, J., and Houlsby, N. (2021). An image is worth 16x16 words: Transformers for image recognition at scale. In *International Conference on Learning Representations (ICLR)*.
- Foret, P., Kleiner, A., Mobahi, H., and Neyshabur, B. (2021). Sharpness-aware minimization for efficiently improving generalization. In *ICLR*.
- Ganin, Y., Ustinova, E., Ajakan, H., Germain, P., Larochelle, H., Laviolette, F., Marchand, M., and Lempitsky, V. (2016). Domain-adversarial training of neural networks. *Journal of Machine Learning Research*, 17(59):1–35.
- Geirhos, R., Rubisch, P., Michaelis, C., Bethge, M., Wichmann, F. A., and Brendel, W. (2019). ImageNet-trained CNNs are biased towards texture; increasing shape bias improves accuracy and robustness. In *ICLR*.

- Gretton, A., Borgwardt, K. M., Rasch, M. J., Schölkopf, B., and Smola, A. (2012). A kernel two-sample test. *Journal of Machine Learning Research*, 13(25):723–773.
- He, K., Zhang, X., Ren, S., and Sun, J. (2016). Deep residual learning for image recognition. In *IEEE Conference on Computer Vision and Pattern Recognition (CVPR)*, pages 770–778.
- Hendrycks, D. and Gimpel, K. (2017). A baseline for detecting misclassified and out-of-distribution examples in neural networks. In *ICLR*.
- Huang, X. and Belongie, S. (2017). Arbitrary style transfer in real-time with adaptive instance normalization. In *IEEE International Conference on Computer Vision (ICCV)*, pages 1501–1510.
- Ilharco, G., Wortsman, M., Wightman, R., Gordon, C., Carlini, N., Taori, R., Dave, A., Shankar, V., Namkoong, H., Miller, J., Hajishirzi, H., Farhadi, A., and Schmidt, L. (2021). OpenCLIP. Zenodo. <https://doi.org/10.5281/zenodo.5143773>.
- Krizhevsky, A. (2009). Learning multiple layers of features from tiny images. Technical report, University of Toronto.
- Lee, K., Lee, K., Lee, H., and Shin, J. (2018). A simple unified framework for detecting out-of-distribution samples and adversarial attacks. In *Advances in Neural Information Processing Systems (NeurIPS)*, volume 31.
- Li, D., Yang, Y., Song, Y.-Z., and Hospedales, T. M. (2017). Deeper, broader and artier domain generalization. In *ICCV*, pages 5542–5550.
- Liu, W., Wang, X., Owens, J. D., and Li, Y. (2020). Energy-based out-of-distribution detection. In *NeurIPS*, volume 33.
- Long, M., Cao, Y., Wang, J., and Jordan, M. I. (2015). Learning transferable features with deep adaptation networks. In *International Conference on Machine Learning (ICML)*, pages 97–105.
- Long, M., Cao, Z., Wang, J., and Jordan, M. I. (2018). Conditional adversarial domain adaptation. In *NeurIPS*, volume 31.
- Loshchilov, I. and Hutter, F. (2019). Decoupled weight decay regularization. In *ICLR*.
- Radford, A., Kim, J. W., Hallacy, C., Ramesh, A., Goh, G., Agarwal, S., Sastry, G., Askell, A., Mishkin, P., Clark, J., Krueger, G., and Sutskever, I. (2021). Learning transferable visual models from natural language supervision. In *ICML*, pages 8748–8763.
- van der Maaten, L. and Hinton, G. (2008). Visualizing data using t-SNE. *Journal of Machine Learning Research*, 9(86):2579–2605.
- Vaze, S., Han, K., Vedaldi, A., and Zisserman, A. (2022). Open-set recognition: A good closed-set classifier is all you need? In *ICLR*.
- Verma, V., Lamb, A., Beckham, C., Najafi, A., Mitliagkas, I., Lopez-Paz, D., and Bengio, Y. (2019). Manifold mixup: Better representations by interpolating hidden states. In *ICML*, pages 6438–6447.
- Wilson, E. B. (1927). Probable inference, the law of succession, and statistical inference. *Journal of the American Statistical Association*, 22(158):209–212.
- Zhou, D.-W., Ye, H.-J., and Zhan, D.-C. (2021). Learning placeholders for open-set recognition. In *CVPR*, pages 4401–4410.